

**This assignment will not be graded and is only for practice.**

## Level 1

1) The null space of a matrix  $A_{4 \times 3}$  is

**1 point**

- ☐ the subspace  $W = \{x \in \mathbb{R}^4 \mid Ax = 0\}$  of  $\mathbb{R}^4$ .
- ☐ the subspace  $W = \{x \in \mathbb{R}^3 \mid Ax = 0\}$  of  $\mathbb{R}^3$ .
- ☐ the subspace  $W = \{x \in \mathbb{R}^2 \mid Ax = 0\}$  of  $\mathbb{R}^2$ .
- ☐ the first column of the matrix  $A_{4 \times 3}$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

the subspace  $W = \{x \in \mathbb{R}^3 \mid Ax = 0\}$  of  $\mathbb{R}^3$ .

2) Nullity of a matrix  $A_{3 \times 4}$  is

**1 point**

- ☐ 3
- ☐ 4

the number of independent variables in the system of linear equations  $Ax = 0$ ,

- ☐ where  $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ .

the number of dependent variables in the system of linear equations  $Ax = 0$ ,

- ☐ where  $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

the number of independent variables in the system of linear equations  $Ax = 0$ , where  $x =$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}.$$

Consider the system of linear equations  $Ax = 0$ , where  $x = [x_1 \ x_2 \ x_3]^T$  and  $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ .

Use the above information to answer questions 3, 4 and 5.

3) Choose the correct statement from the following options.

**1 point**

- ☐  $x_1, x_2$  and  $x_3$  all are dependent variables.
- ☐  $x_2$  is the only dependent variable.
- ☐  $x_1$  and  $x_3$  are the dependent variables and  $x_2$  is the only independent variable.
- ☐  $x_1, x_2$  and  $x_3$  all are independent variables.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x_1$  and  $x_3$  are the dependent variables and  $x_2$  is the only independent variable.

4) Which one of the following vector spaces represents the null space of  $A$ ?

**1 point**

- ☐  $\{(t_1, 0, t_2) \mid t_1, t_2 \in \mathbb{R}\}$
- ☐  $\{(0, t, 0) \mid t \in \mathbb{R}\}$
- ☐  $\{(t_1, t_2, 0) \mid t_1, t_2 \in \mathbb{R}\}$
- ☐  $\{(0, 0, t) \mid t \in \mathbb{R}\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(0, t, 0) \mid t \in \mathbb{R}\}$

5) What will be the nullity of  $A$ ?

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No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

**1 point**

6) Find the nullity of the matrix  $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 4 & 6 \end{bmatrix}$ .

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No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

**1 point**

7) Let  $S$  denote the set of solutions of the homogeneous system of linear equations  $Ax = 0$ . Which of the following statements is (are) true? **1 point**

- ☐ If  $x_1$  and  $x_2$  are in  $S$ , then  $x_1 - x_2$  is in  $S$ .
- ☐ If  $x$  is in  $S$ , then  $cx$  is in  $S$ , for any  $c \in \mathbb{R}$ .
- ☐ If  $x_1$  and  $x_2$  are in  $S$ , then  $\alpha x_1 + \beta x_2$  is in  $S$ , for any  $\alpha, \beta \in \mathbb{R}$ .
- ☐ If  $x$  is in  $S$ , then  $A^n x = 0$  for any  $n \in \mathbb{N} \setminus \{0\}$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

If  $x_1$  and  $x_2$  are in  $S$ , then  $x_1 - x_2$  is in  $S$ .

If  $x$  is in  $S$ , then  $cx$  is in  $S$ , for any  $c \in \mathbb{R}$ .

If  $x_1$  and  $x_2$  are in  $S$ , then  $\alpha x_1 + \beta x_2$  is in  $S$ , for any  $\alpha, \beta \in \mathbb{R}$ .

If  $x$  is in  $S$ , then  $A^n x = 0$  for any  $n \in \mathbb{N} \setminus \{0\}$ .

8) Find out the value of  $a$  for which the matrix  $\begin{bmatrix} 1 & 2 \\ 3 & a \end{bmatrix}$  has nullity 1.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

**1 point**

**Level 2**

9) Consider the coefficient matrix  $A$  of the following system of linear equations:

**1 point**

$$x_1 + x_2 + x_4 = 0$$

$$x_2 + x_3 = 0$$

$$x_1 - x_3 + x_4 = 0$$

Which one of the following vector spaces represents the null space of  $A$ ?

- ☐  $\{(t_1 + t_2, t_1, t_1, t_2) \mid t_1, t_2 \in \mathbb{R}\}$
- ☐  $\{(t_1 - t_2, -t_1, t_1, t_2) \mid t_1, t_2 \in \mathbb{R}\}$
- ☐  $\{(t_1 + t_2, -t_1, t_1, t_2) \mid t_1, t_2 \in \mathbb{R}\}$
- ☐  $\{(t_1 - t_2, -t_1, t_1, -t_2) \mid t_1, t_2 \in \mathbb{R}\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\{(t_1 - t_2, -t_1, t_1, t_2) \mid t_1, t_2 \in \mathbb{R}\}$$

10) Find the nullity of the matrix  $A$ , where  $A = [a_{ij}]$  is of order  $3 \times 3$  and  $a_{ij} = \min\{i, j\}$ ,  $i, j = 1, 2, 3$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

**1 point**

11) Choose the correct set of options from the following.

**1 point**

- ☐ The nullity of a non-zero scalar matrix of order 3 must be 3.
- ☐ The nullity of a non-zero scalar matrix of order 3 must be 0.
- ☐ The nullity of a non-zero diagonal matrix of order 3 must be 3.
- ☐ The nullity of a non-zero diagonal matrix of order 3 can be at most 2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The nullity of a non-zero scalar matrix of order 3 must be 0.

The nullity of a non-zero diagonal matrix of order 3 can be at most 2.

12) Consider the system  $Ax = b$ , where  $A = \begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix}$ . The number of solutions of when  $b = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$  is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

**1 point**

Your score is: 0/12